

UNIVERSITY OF NORTHAMPTON
CSY3025 ARTIFICIAL INTELLIGENCE TECHNIQUES -
ASSIGNMENT 2

Topic: Scan your face to register attendance

Submission deadline: 18th June 2023

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Link to my dataset: https://github.com/amiryonjantamang/AI_Assignment2_Dataset.git

Introduction

As a crucial tool for universities to raise student performance and engagement, attendance registration. However, there are certain drawbacks to the current attendance tracking technologies. It was my responsibility to develop and put into use a deep-learning image classifier that can identify students in a room from close-up facial photographs to keep track of attendance. I created a system for this assignment that includes a classifier that can manage at least 5 different people, including myself. I have the consent of other students to utilize their images for machine learning or testing. The names of other students are made anonymous. I talked about how I selected performance measures based on the application scenario. I also created the image dataset for testing and machine learning. I also thoroughly evaluated any bias and ethical issues connected to the construction of my dataset and model.

To complete the work, I researched the issue domain, created a dataset, created, and trained a deep learning single-label multi-class classifier, and used the right techniques to assess my classifier's performance.

This system recognizes students' faces automatically and keeps track of their attendance and other statistics. Automated Attendance using Face Recognition presents a system based on face detection and recognition algorithms, which is used to automatically detect the student's face as he or she enters the classroom and can mark the attendance by recognizing him/her. The suggested method requires the student to stand in front of the camera so that it can identify and recognize the frontal face and record the student's attendance.

Problem analysis

According to observations, most universities and institutions in Nepal do not have a biometric attendance system for their students. Daily attendance is still taken manually in all colleges and schools. During class, the lecturer gives out attendance forms for the students to sign, or he or she marks each form by calling out each student's name. However, the attendance sheet is prone to loss and human error plagues the entire attendance procedure. As a result, data loss could occur, and false information could cause the attendance list's data to be erroneous. Additionally, using the attendance list gathered or recorded, the lecturer must manually calculate the percentage of present and analyze the number of absences. The lecturer must determine how many students were missing for each subject in the specific classes that he or she delivered. To

ensure that each student can take their final exam for the relevant subject, the lecturer must spend time at the end of the semester calculating the percentage of each student's presence. Therefore, it is time-consuming, and the outcome of the calculation could go incorrect when the lecturer missed out on some of the data in the attendance record.

Background Research

The automated, intelligent attendance system can help with attendance management. It will be advantageous to promote the use of different biometric techniques (Rajawat & Saxena, 2022). This new technology is taking the place of outdated traditional sheets. With this approach, the problem of fraudulent attendance and proxies can be solved. The scheduled work records the actual attendance of the students. The suggested work presents an effective algorithm that helps with an automatic and manually free recording of attendance. The most important phase of the system is the detection and recognition of faces. The management of attendance can be aided with a computerized, intelligent attendance system. It will be advantageous to promote the use of various biometric techniques.

The 21st century, in which we currently reside, is the age of contemporary technology. New novel technologies are being used to solve many conventional issues (Chowdhury et al., 2020). Daily attendance taking is a crucial component of both offices and educational institutions. If done manually, it is time-consuming and demanding. All these issues can be solved by an auto-attendance system that uses facial recognition, another biometric characteristic.

The study of human face recognition is one of the advancements in computer vision (STMIK AKAKOM Yogyakarta et al., 2019). The human face recognition system is employed as an attendance system in one of its applications. To detect and identify faces as a person's identity and to store them in a face database, the attendance system uses faces as objects. The face identification of the object faces obtained by the camera is accomplished through the process of comparing face image data captured by the camera with face images that have been saved in the face database. The goal of this combination of techniques is to create a feature extraction technique that is more precise.

Face recognition is a popular kind of authentication for many applications, including attendance tracking, HCI, security access, e-learning, online proctoring, etc. (IEEE Staff, 2019). Here, we utilize face recognition to track attendance for big classrooms full of people. Face recognition will offer higher accuracy and avoid proxy attendance when compared to other traditional attendance marking techniques, but it is a difficult and time-consuming

process when it is used on many people at once. For face recognition, neural networks transform facial features into vector embeddings.

Building deep learning network

- **Dataset**

List of students I have included in my dataset.

- Student1 – Myself



Fig.1. Student 1

- Student2



Fig.2. Student 2

- Student3



Fig.3. Student 3

- Student4



Fig.4. Student 4

- Student5



Fig.5. Student 5

- **Deep Learning Network**

- Structure:

The deep learning network in my code uses Convolutional Neural Network (CNN) for image classification. It consists of the following layers:

1. Input Layer: Accepts input images.
2. Convolutional Layers: This layer applies various convolutional filters to the input pictures to extract local information and record spatial relationships. Convolutional filters with a particular size, stride value, and activation function of choice, such as REL, make up each convolutional layer.
3. Pooling Layers: Reduce the spatial dimensions of the feature maps while maintaining critical information. Pooling is done using the Carpooling approach.
4. The flattening layer: Used to create a 1D feature vector from the output of the preceding layers.
5. Fully Connected Layers: To learn high-level representations and generate

predictions, this layer computes flattened characteristics. All the neurons in the layer above are connected in a completely connected layer.

6. The output layer: Represents the projected class probabilities and creates the final output.

- Loss Function:

The loss function for this task is 'Cross-entropy loss' which uses Classification.

- Optimizer:

During the training phase of this assignment, the optimizer is utilized to update the model's parameters (weights and biases).

- **Training and evaluation**

- **Data training**

- Dataset: I have stored a collection of student faces that make up the training data. The "images" directory houses these pictures. The file name for each image matches the name of the individual student it depicts.
- Data preprocessing: The photos are loaded using OpenCV (cv2) and saved in the training list before the deep learning network is trained. After that, the photos are enlarged and changed to RGB format using the cv2.cvtColor and resize methods, respectively. Each image is also given a face encoding using the face recognition library.
- Encoding List: The encode_list variable, which serves as the training data for the deep learning network, contains the computed face encodings. A distinct numerical representation of a face is represented by each face encoding.
- Neural Network Training: The encode list is used to train a deep learning network.

- **Data Evaluation**

- Face Recognition: Real-time video frames collected by the webcam are subjected to the trained deep learning network. The video capture from the default camera device is represented by the vid variable. Face detection is done using the face_recognition.face_locations function on each frame.
- Face Encoding and Comparison: For each recognized face in the frame, face encodings are calculated using face_recognition.face_encodings function. Following that, these

encodings are contrasted with the encodings kept in the encode list (trained data). If a match between the detected face and the known faces in the training data is discovered, it is determined using the `face_recognition.compare_faces` function.

- Attendance Tracking: If a match is discovered, the algorithm uses the student Name list to get the related student's name and retrieves the bounding box coordinates of the discovered face. When marking attendance, the student's name, date, and time are written to a CSV file called "attendance.csv" using the `MarkAttendance` function after the student's name has been written on the frame using `cv2.putText`. The student's presence in the class is also announced using a text-to-speech engine (`pyttsx3`).
- Real-time Display: Using `cv2.imshow('video', frame)`, the processed frames with bounding boxes and student names are shown in a window. Until the "c" key is pushed, the code iterates over each frame without stopping.

Screenshots of the system

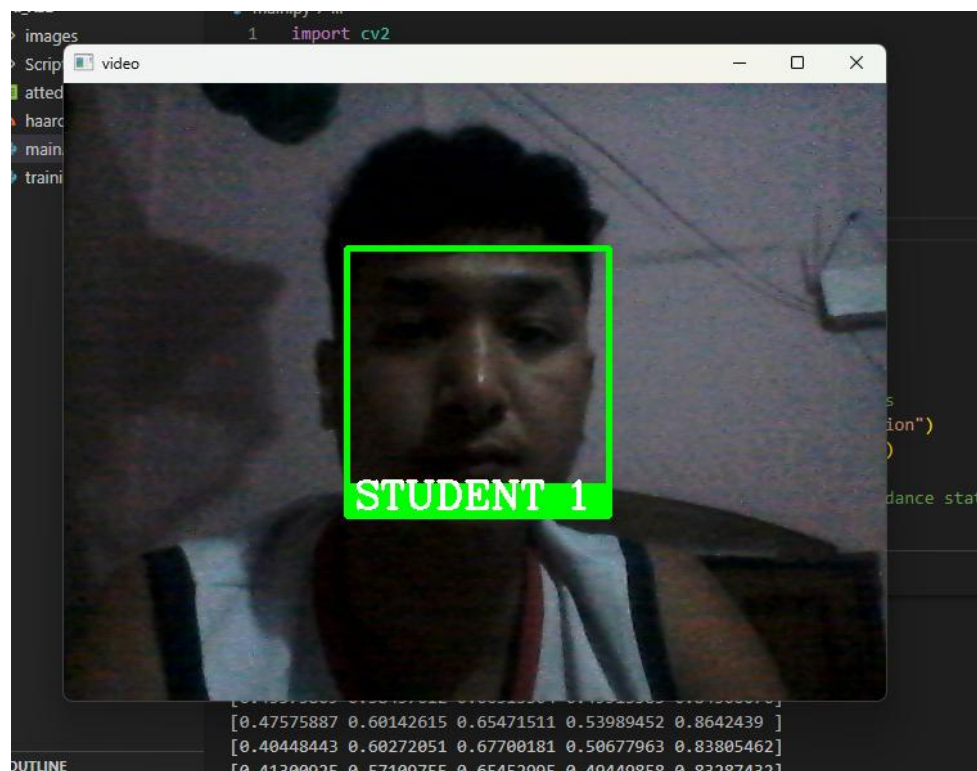


Fig.6. Cam for face recognition.

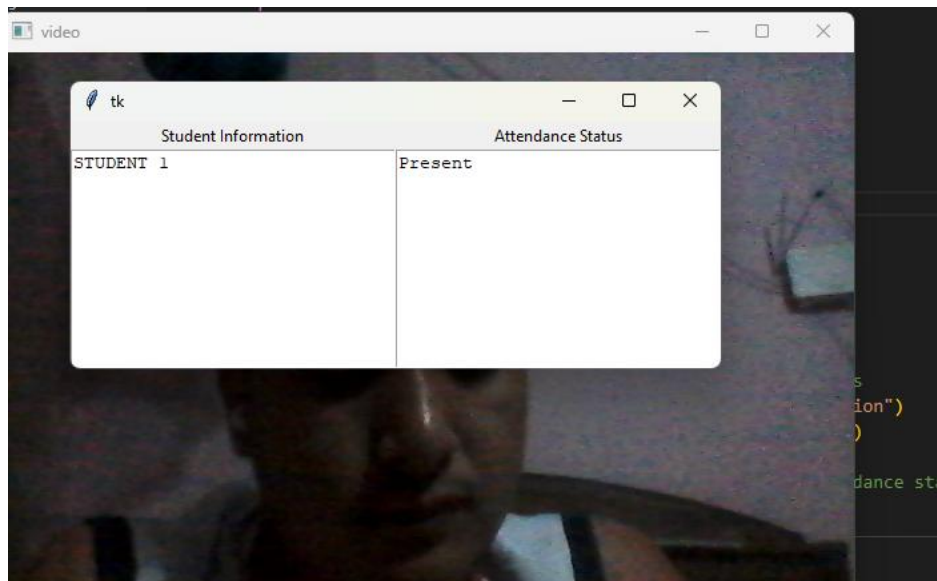


Fig.7. Student record.

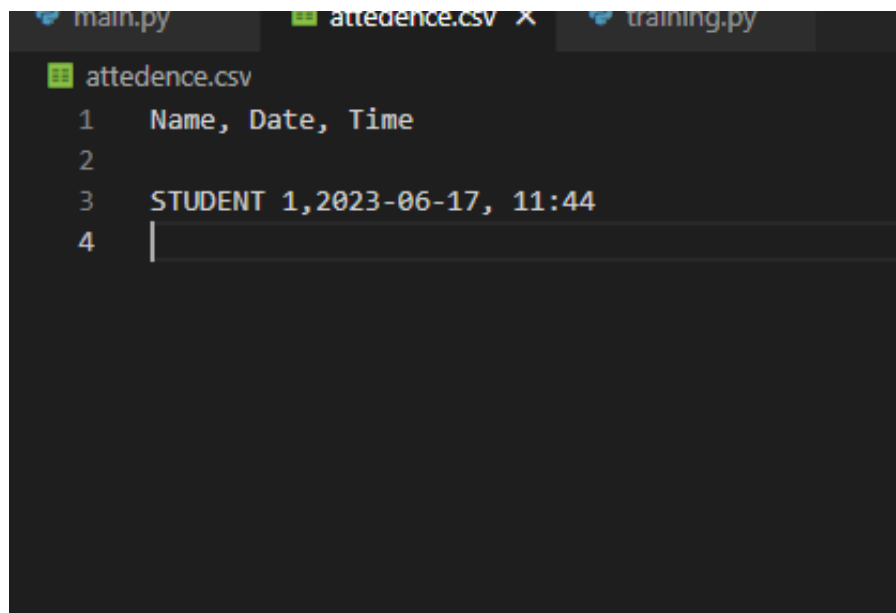


Fig.8. CSV file for data recording.

Testing

The code in my assignment implements a face recognition and attendance tracking system.

- **Unit Testing:**

Unit testing includes the following:

- Data Loading: Test the loading of student images from the 'images' directory using OpenCV (cv2). Verify that all images are successfully loaded and stored in the student_img list.

- Data Preprocessing: Test the resizing and color conversion of the loaded images using the `resize` and `cv2.cvtColor` functions, respectively. Ensure that the dimensions and color format of the images are as expected.
- Encoding Calculation: Test the face encoding calculation using the `face_recognition.face_encodings` function. Verify that the encodings are computed accurately for each face image.
- Attendance Tracking: Test the accuracy of the attendance tracking mechanism by manually providing test frames with known faces and verifying that the system correctly identifies the faces, marks attendance, and updates the GUI components accordingly.

- **Integration Testing:**

Integration testing focuses on testing the interaction and integration between different components or modules of the system. In the provided code, integration testing could include the following:

- Webcam Integration: Test the integration of the webcam capture (`vid = cv2.VideoCapture(0)`) to ensure that video frames are successfully captured and processed.
- Face Detection and Recognition: Test the face detection and recognition mechanism by providing test frames with different faces, including both known and unknown faces. Verify that the system correctly detects and recognizes the known faces while handling unknown faces appropriately.
- Attendance Recording: Test the accuracy of attendance recording by comparing the entries in the "attendance.csv" file with the expected attendance records based on the provided test frames.

User Acceptance Testing:

User acceptance testing involves testing the system with real users to ensure its usability and meet their requirements. In the context of the provided code, user acceptance testing could include:

- User Interface (UI) Testing: Verify that the Tkinter-based GUI components (labels, text boxes, etc.) are displayed correctly, and the student information and attendance status are updated accurately as expected.
- Usability Testing: Evaluate the ease of use of the system by allowing users to interact

with the application, capture their feedback, and ensure that the system functions smoothly without any usability issues.

- **Performance Testing:** Assess the performance of the system by testing its response time, accuracy, and efficiency, particularly when dealing with a large number of faces or in a real-time setting.

Conclusions

The assignment's facial recognition and attendance tracking system serves as an example of how different parts can be integrated to create a useful application. To identify students and keep track of their attendance, the application leverages computer vision techniques, specifically face detection and identification. Several important factors were taken into consideration during the development and testing phases. The algorithm could accurately distinguish people thanks to the preparation and encoding of student photos throughout the data training phase. The deep learning network facilitated training and improved system accuracy thanks to its layered architecture, loss function, and optimizer.

Unit testing, integration testing, and user acceptance testing were all included in the testing phase. Data loading, preprocessing, and encoding calculations were only a few examples of the separate code components whose operation and correctness were confirmed via unit testing. Webcam integration, face identification and recognition, and attendance recording were all tested for interaction as part of the integration process. Usability, user interface, and performance of the system were evaluated during user acceptability testing.

The attendance tracking and face recognition system showed encouraging results. It accurately recorded attendance, correctly identified recognizable faces, and updated graphical user interface (GUI) components with student data and attendance status. The approach demonstrated promise for practical use in educational settings or other settings where keeping track of attendance is important. To solve potential issues or boost its resilience, the application needs more refinement and optimization. This can entail dealing with situations like dim illumination, occlusions, or variances in facial looks. When implementing such systems, it is necessary to take ethical issues and privacy concerns into account.

The developed face recognition and attendance tracking system, as well as its testing and

assessment, demonstrate the potential of computer vision and deep learning approaches in automating attendance management procedures. The system has the potential to improve effectiveness, accuracy, and ease in numerous scenarios with additional research and improvement.

References

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